

Appendix E: Packaging Compliance Self-Audit Form

Void space, hazardous substances, recyclability — item-by-item check Applies to: all sellers shipping packaged goods to the EU

How to Use

Check each item against your current packaging. Any item that fails requires corrective action.

Part 1: Void Space Ratio Check

Calculation Formula

Void Space Ratio = (Packaging Volume – Product Net Volume) ÷ Packaging Volume × 100%

From 2030: E-commerce parcel void space must be ≤ 50%

Steps

1. **Measure packaging internal dimensions** (L × W × H)
2. **Calculate packaging volume** = L × W × H
3. **Measure product net volume** (including inner packaging)
4. **Calculate void space ratio**
5. **Compare against standard**

Self-Audit Table

Check Item	Your Value	Standard	Pass?
Total packaging volume (cm³)	[fill in]	—	—
Product net volume (cm³)	[fill in]	—	—
Void space ratio (%)	[fill in]	≤ 50%	■
Uses bubble wrap / foam filling	■	Counts toward void	—
Extra packaging layers	■	Violation	—
Double-wall cartons / hidden false bottoms	■	Violation	—

Corrective Actions

- ■ Reduce outer box dimensions
- ■ Decrease filler material volume
- ■ Switch to more compact packaging design
- ■ Consider foldable / compressible packaging options

Part 2: Restricted Substances Check

Limit Standards

Test Item	Limit	Your Result	Pass?
Heavy metals (Pb + Cd + Hg + Cr VI) total	≤ 100mg/kg	[fill in]	■
Individual PFAS (quantitative)	≤ 25ppb	[fill in]	■
Total individual PFAS sum	≤ 250ppb	[fill in]	■
Polymer PFAS total fluorine	≤ 50ppm	[fill in]	■

High-Risk Materials Self-Check

Material Type	Used?	Risk Level	Notes
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Bubble mailers / bubble wrap	■	■ High	Common PFAS contamination
Foam inserts / EPE pearl cotton	■	■ High	Common PFAS contamination
Coated paper / laminated bags	■	■ Medium	Requires testing
Plastic carrier bags (< 15µm)	■	■ High	Banned from 2030
Food-contact packaging	■	■ High	No transition period
Fresh produce shrink wrap	■	■ High	Banned from 2030

Testing Recommendations

- ■ Arrange packaging material testing with a third-party lab
- ■ Prioritise bubble mailers and foam materials (highest PFAS risk)
- ■ Retain test reports as compliance evidence
- ■ Food-contact packaging requires additional PFAS testing

Part 3: Recyclability Check

Packaging Recyclability Grades

Packaging Type	Estimated Grade	Needs Correction?
-----	:---:	:---:
Cardboard / corrugated board	A (Highly recyclable)	■
Plastic film (PE)	B (Moderately recyclable)	■
Composite / laminated bags	C (Poorly recyclable)	■ ■■
Coated paper	C (Poorly recyclable)	■ ■■
Foam plastics	C (Poorly recyclable)	■ ■■
Glass bottles	A (Highly recyclable)	■
Metal cans	A (Highly recyclable) ■	

Correction Directions

- ■ Replace C-grade packaging with A/B-grade materials
- ■ Reduce use of composite / laminated materials
- ■ Prioritise single-material packaging
- ■ Consider recycled-content packaging options

Part 4: Labeling Check

Required Labels

Label Type	Present?	Notes
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Recycling pictogram (standardised)	■	PPWR uniform standard
Packaging material code	■	e.g., PET, HDPE, paper
Producer information	■	Company name / address
EPR registration number	■	Country-specific number
Country of origin	■	e.g., Made in China
Material / ingredient breakdown	■	As required by regulation

Conclusion

Your packaging is PPWR-compliant only when all items above pass.

If any item fails, develop a correction plan following the guidance in Chapter 3 of this guide.

This form is based on the EU PPWR Regulation text and official implementation guidelines. Testing standards and limits may be updated; always verify with official sources.